

MEETING MINUTES

Project: ELEN4000A/4011A Design II — Topic 2: Design and Simulation of a Nano-Grid for African Electrification

Supervisor: Mr Mohammed Raees Dangor

Date: Monday, 20 July 2026

Type: First supervisor meeting, Week 1

1. Background / Context Given by Supervisor

- Dr Dangor shared the history behind the project brief, drawn from his own industry experience at Ecopower/Pico Power, including a failed early deployment ("the Dixieland project") — a system that was over-engineered and built without fully understanding the target community.
- Global context: in 2010, roughly 1 billion people worldwide lacked access to electricity (the reason People Power/Pico Power was founded). Students are to find and report the current updated figure for this in their report.
- Illustration used: satellite night-light imagery of Africa shows vast unlit regions; when entire European countries are overlaid onto the African continent, they can fit into gaps with no high-voltage (HV) line contact at all — demonstrating how sparse HV transmission infrastructure is across much of Africa.
- Centralised generation (generation → transmission → distribution, as used in Europe/SA) does not work well in Africa due to rugged terrain and sparse, scattered populations — hence the case for distributed/off-grid generation models instead.
- South African electrification history: rose from ~60% in 1994 to ~90% today (Eskom, source: General Household Survey). However, the 90% figure is a "false narrative" if taken alone — it doesn't capture other electrification dimensions like availability, cost, and safety of the connection.

2. Why the ≥20km-from-HV-line Requirement

- This specification exists to force a genuine off-grid engineering problem: if a settlement is close to the grid, the correct (and cheaper) engineering answer is simply to extend the grid, not design a standalone system.
- The brief also assumes zero existing electrification at the chosen site ("from the ground up") — students are not extending or connecting to any existing infrastructure.

3. ESMAP Multi-Tier Framework (MTF)

- Central to the project — students must justify which tier of electrification they are designing for.
 - Tier 1: lighting, cell phone charging

- Tier 2: TV / home entertainment
- Tier 3: refrigeration
- Tier 4: heating
- Tier 5: extreme heating and cooking
- Each tier corresponds to an associated wattage/output level.
- Reference: Dr Dangor's own Master's thesis (available on WiredSpace, Wits repository) covers the MTF and inspired his PhD work on measuring "poverty" more broadly (energy poverty → digital divide → AI divide as a progression).

4. Technology and Architecture

- Two main architecture options: centralised vs distributed generation (this maps to the CBO's brief language too).
- Dr Dangor's own Master's thesis (WiredSpace) has a dedicated section comparing centralised vs distributed generation — required reading.
- Also referenced: his SAUPEC 2019 conference paper, titled around "Personal Consumer Grid" / PicoGrid, describing how Pico Power's system and models work — required reading.
- Pico Power itself runs two models: the CDM Core (Centralised Distribution Model — single central point/box with storage and tech at a high point, e.g. a hill, articulated out to houses) and a modular/distributed system.
- Students should research named competitor/reference technologies in the project brief (e.g. Solar Turtle, and others listed) — note: a couple of names mentioned verbally in the meeting were unclear in the recording; cross-check the actual written project brief document for the exact competitor list rather than relying on the audio.
- Grid size terminology (pico / nano / micro grid) varies across the literature — students should mention a few definitions in the report but pick and justify the one that fits their design. Not strictly limited to "nano" scale, though micro-grid scale was discouraged as likely too large for this project's scope.
- A past student (Fatima Mulla — highest mark in last year's cohort) has a strong section comparing pico/nano/micro grids; Dr Dangor offered to check if she'd share her report with the group for reference.

5. Business / Socio-economic Model

- Just as important as the technical design.
- Cautionary example given: a peer-to-peer energy trading model his team once proposed failed because it assumed community members could pay each other in livestock/goods without verifying that most households actually owned livestock — illustrating the risk of designing without understanding the real socio-economic context of the community.
- Students must think through how the community will pay for/sustain the system (this maps directly onto the CBO's "Revenue Model" requirement).

6. Simulation Guidance

- Dr Dangor stated that dedicated simulation software is not mandatory — simulation is fundamentally about proving/validating a design concept (a "dress rehearsal" for the idea), and this can be done by clearly specifying parameters and design assumptions to an AI tool to generate outputs, rather than requiring a specific traditional simulation package.
- **Note:** this appears to differ from the CBO's stated requirement that simulation software be student's choice but supervisor-approved, and from the general Course AI Policy restricting AI use to grammar/clarity editing only, not idea generation or original analysis. Recommend the team gets explicit written clarification from Dr Dangor on exactly what is permitted here, given the stricter course-wide AI policy in the CBO — to avoid conflict between supervisor guidance and formal course policy in the eventual write-up.

7. Expectations of Students

- This is a client-style engagement: Dr Dangor is acting as "the client" solving his own problem, not spoon-feeding weekly tasks.
- Students are expected to independently plan their weekly tasks, bring this plan to him, and he will guide/course-correct based on what they propose — not the reverse.
- Emphasis throughout: the engineer's core value is in thinking and scoping the problem, not just execution — AI can be leveraged heavily for execution/detail work once the underlying design thinking is the student's own.
- **Deliverable requested this week:** the group must compile a list of everything they've decided needs to be done in Week 1, send it as a PDF to Dr Dangor via the WhatsApp group.

8. Action Items Raised in the Meeting

- Find and confirm a specific unelectrified site (≥ 20 km from HV line), using Google Earth.
- Review relevant literature/papers, including Dr Dangor's own Master's thesis and SAUPEC 2019 paper.
- Investigate "energy poverty" and energy access literature (ESMAP MTF).
- Research competitor technologies and business models (Solar Turtle, Pico Power's CDM Core vs modular model, others listed in the brief).
- WhatsApp group created for ongoing communication with supervisor.